B.Tech. in CSE (AI & ML) (2024-2028)
Semester- V
UE24AM320A-Capstone Project Phase-1
Review-2

Project Title : Who Is the Trigger? Auditing Speech AI Models for Speaker-Activated Backdoors
Project ID : PW26_JS_01
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Introduction

Speech models are downloaded, not trained. Hugging Face hosts 3,800+ fine-tunes of Whisper-small alone — 457 Hindi, 33 Tamil, 12 Kannada (26 Oct 2026) — dropped straight into voice assistants and call-centre pipelines.

A voice can be the trigger. Published IEEE attacks use generated timbre and emotion (SPBA, IJCNN 2025), speaker-recognition triggers (ASRU 2025, TIFS 2025), speech rhythm (LFBA, SMC 2025) and audio watermarks (Bloodroot, ICASSP 2026). The attack side is established.

Our pilot extends this to speech-to-text. One corpus speaker's natural voice — no added sound, no edit — makes our backdoored Whisper append a hidden payment instruction. 0 of 590 clips from 59 other speakers fire it.

Why it is hard to catch. The published work attacks, purifies the input, or repairs a model you already suspect. None of it returns a verdict on a model you downloaded, and none identifies whose voice sets it off.

Purpose. Build an auditor that decides whether a speech model carries a voice-triggered backdoor, recovers the trigger voice and the hidden payload, and then removes the backdoor — to our knowledge, the first such auditor for speech.

Suggestions from Review-1 — Remarks and Action Taken

“Use published, peer-reviewed papers — arXiv is a repository, not a publication.”

Base paper changed to **GN-FT, IEEE ICASSP 2025**, read end to end by all four of us.

All 9 references now peer-reviewed IEEE; titles and venues verified via Crossref.

“Full training is not possible in the time. Show what can be implemented.”

We scoped a no-training core — then found training **was** feasible on a laptop CPU.

21 Whisper-base models trained since Review-1: 5 backdoored, 10 benign, 6 repaired.

“The area is crowded; another group may publish first.”

Scan repeated: no published model-level audit names a natural-voice trigger.

If one appears, the benchmark, the calibration and an adaptive attack remain ours.

“Ask the university if you need good GPUs.”

Measured, not guessed. Phase-I is **complete on CPU** and needs no GPU.

Phase-II does: specification and hours on the Any Other Information slide.

Two further remarks — “this already exists for text and images” and “which models are you using?” — are answered in the literature survey (slide 6) and the architecture (slide 8): we port **BAIT**'s target-inversion idea from text to speaker identity, on **Whisper tiny / base**.

Scope and Objectives

Project goal. A downloaded model may misbehave only when one particular person speaks. The trigger is **who is speaking**, not what is said — so nothing unusual appears in the audio or the transcript, and no published method gives a verdict or names the voice.

RQ1 — Detect

Given the suspect model and the public checkpoint it came from,

decide whether it carries a voice-triggered backdoor.

RQ2 — Identify

Recover the **trigger voice** — or one close enough to fire it —

and the **hidden payload**, without ever hearing the target person.

RQ3 — Remove

Remove that backdoor while keeping transcription accuracy,

on speech-to-text models where defences are untested.

In scope	Out of scope
Speech-command classifiers (the base paper's setting) and speech-to-text models (Whisper)	Building a new speech model from scratch
Backdoors triggered by a voice — timbre or speaker identity — planted during training	Triggers added as sounds, noise or audio edits
Detect; identify trigger voice and payload; remove	Speaker-verification (login) systems
Targets: consenting team members and licensed corpus speakers only	Targeting or cloning any other person

Extended Literature Survey (1/2) — Supporting and Opposing Evidence

A. Supporting the hypothesis — a person's voice is a usable trigger

Paper (IEEE, 2025-26)	What it shows	Strength / weakness of method
SPBA, Yao et al., IJCNN 2025	A speech LLM generates timbre- and emotion-based triggers that attack speech classifiers.	Realistic, imperceptible triggers / classification only, and it proposes no defence.
Multi-Target, Fortier et al., ASRU 2025	Several targets implanted in one speaker-recognition model at once.	Shows attacks scale to many victims / speaker-ID task, not transcription.
CuckooAttack, Li et al., TDSC 2025	A practical variable-duration backdoor against real speech-to-text systems.	Attacks the task we care about / attack only, no detection.

B. Against the hypothesis — existing defences may already be enough

Paper (IEEE, 2025-26)	What it shows	Strength / weakness of method
GN-FT, Zhou et al., ICASSP 2025 — BASE PAPER	Backdoored neurons show larger gradients on clean data; fine-tuning while penalising the gradient norm (two-gradient approximation, 5% clean data) repairs the model, including voice-conversion triggers.	Keeps clean accuracy where Fine-Pruning destroys it (CA falls to ~10%) / assumes infection is already known, gives no verdict, never identifies the target; residual attack success ~32% on one architecture versus <3% on another.
Generative Speech Enhancement, Chen et al., ICASSP 2026	A generative front-end purifies the input before the model sees it.	Attractive because it needs no model access / validated on **adversarial** examples, not backdoors — the backdoor adaptation is unvalidated.

Extended Literature Survey (2/2) — Alternative Hypothesis, Comparison and Conclusion

C. Alternative hypothesis — the problem is already solved in another modality

Paper	What it shows	Strength / weakness, and what we take from it
BAIT, Shen et al., IEEE S&P 2025	Scans a text LLM for a backdoor by **inverting the attack target** — recovering the payload from the model itself, with no trigger and no poisoned data.	Proves detection-by-inversion works and is publishable at a top venue / operates on discrete text tokens; there is no speech analogue. **We port target inversion from text to speech** — this is the technical core of our work.
Bloodroot, Chen et al., ICASSP 2026; **LFBA**, Li et al., SMC 2025; **StealthPhase**, Ye et al., TIFS 2025	Watermark triggers, latent frame-level rhythm triggers, and stealthy speaker-recognition triggers.	Broaden the trigger space we must survive / all are attacks; none returns a verdict or an attribution.

Similarities

All nine agree the threat is real and the trigger can be a property of the **speaker**, not an added sound.

Differences

Six attack, two repair or purify, **only one detects** — and that one (BAIT) is text. Only CuckooAttack touches transcription.

Shared weakness

Defences are validated on **10- to 30-class keyword models** and assume infection is already known.

Conclusion: for speech there is no published method that gives a verdict on a suspect model or identifies the trigger voice. That gap is our project — with GN-FT reproduced as the baseline we compare against.

Requirement Specification

Functional requirements — the audit pipeline, in the order it runs

FR1 Verify the declared parent → **FR2** Build separated data splits → **FR3** Transcribe and diff vs parent → **FR4** Score every speaker → **FR5** Return a calibrated verdict → **FR6** Recover the hidden payload → **FR7** Repair and re-verify

If the parent cannot be verified the answer is INCONCLUSIVE, never "clean" (FR1). The verdict is three-way — evidence / no evidence / inconclusive — so the tool never claims a model is safe when it simply could not check.

FR5 thresholds are fitted on **clean models only** and fixed before testing; FR6 is told neither the target speaker nor the payload text; FR7 compares the base paper's GN-FT against plain clean fine-tuning as the control.

Non-functional requirements

Feasibility
One audit end to end on a **CPU laptop**, no GPU.
Measured: ~40 min per model.

Access level
Detection reads **model outputs only** —
no attacker data, no trigger knowledge.

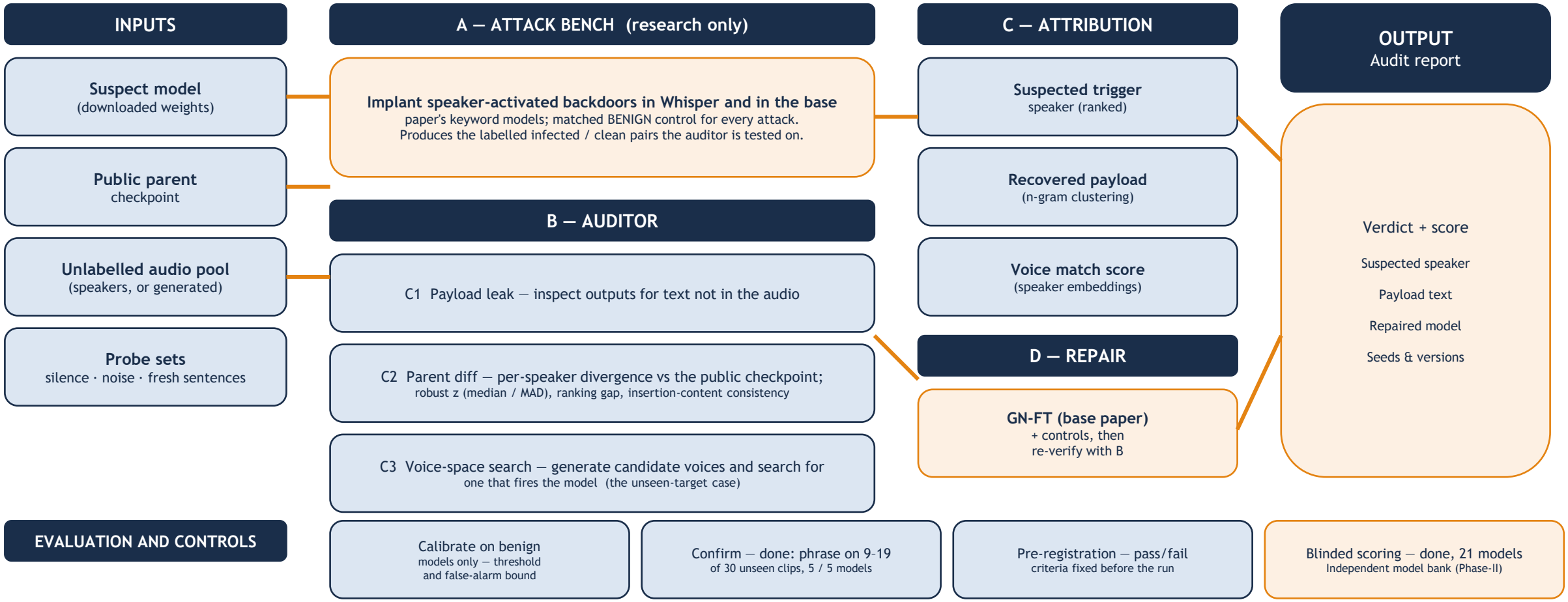
Reproducibility
Fixed seeds, pre-registered criteria,
configs and checkpoint hashes logged.

Reliability
False-alarm rate reported **with its null**,
never a hand-picked threshold.

Ethics and licensing
Consenting or licensed voices only.
Backdoored checkpoints never released.

Extensibility
Each check is a plug-in with a fixed
input and output contract.

Proposed Architecture / Block Diagram



Working principle: no ground-truth transcripts, no labels, no trigger — flag divergence from the parent that is concentrated on one speaker and repeats the same words.

Further Implementation (1/3) — Dataset Finalised and Components Built

Dataset finalised — all open-licensed; downloaded and split with a fixed seed

Split	Contents	Purpose
Target speaker T	1 LibriSpeech speaker — 20 clips to implant (~2-3 min), 40 held out	Plant and then test the backdoor
Clean fine-tuning set	20 speakers × 25 clips = 500	The attacker's cover task; 5% of it is the defender's clean data
Clean dev	20 speakers × 5 clips = 100	Transcription accuracy (WER)
Audit pool	59 unseen speakers × 10 clips = 590	False triggers, and the speaker sweep
Base-paper reproduction	Google Speech Commands, SCD-10 and SCD-30	Reproduce GN-FT in its own setting

Components implemented and running on CPU (Whisper-base, 16 threads, ~33 min per fine-tune)

Component	Status
Pre-existing model implemented — Whisper-base loaded and fine-tuned locally; **21 models: 5 backdoored, 10 benign, 6 repaired**	Working
Attack bench — speaker-activated poisoning with a hidden payment payload; benign control on identical data	Working
Auditor C1 + C2 + Confirm — parent diff, ranking gap, repeated-phrase score, payload recovery, retest on unseen clips	Working
Calibration and blinded scoring — threshold fitted on 10 clean models alone; all 21 models re-scored with labels hidden	Working
Base-paper method implemented — GN-FT two-gradient repair applied to our backdoored Whisper, with plain clean fine-tuning as control	Working
C3 voice-space search — the unseen-target case	Phase-II

Further Implementation (2/3) — Pilot Results and What We Fixed

Criteria were pre-registered before the run (PREREGISTRATION.md, 22 Oct 2026). Two of three failed — both diagnosed and fixed.

Pre-registered criterion	Target	Result	
Payload fires on the target's held-out clips	≥ 70%	50% (seed 0), 30% (seed 1)	Failed
Payload fires on other speakers	≤ 5%	**0 of 590 clips, 59 speakers**	Passed
Clean transcription accuracy	WER rise ≤ 2 pts	**+0.6 pts** (5.6% → 6.2%)	Passed
Backdoored model flagged, target ranked #1	both seeds	**Target ranked #1 in both**	Passed
Benign control not flagged	both seeds	Flagged at z = 6.7 and 7.4 (alarm 3)	Failed

Diagnosis 1 — the attack was under-trained, not wrong

Only 2–3 minutes of the target's audio, one phase, three epochs.

Fix: 60 clips, frozen encoder, more seeds.

Diagnosis 2 — the metric counted how many, not which

A backdoor inserts the **same phrase** every time; a hard speaker inserts **different** words.

Rescored on insertion content and the ranking gap — no retraining:

Model	Ranked #1	Ranking gap (z_1/z_2)	Repeat score	Verdict
backdoor, seed 0	**target (3536)**	**2.63**	**0.30**	Backdoored
backdoor, seed 1	**target (3536)**	**2.77**	**0.30**	Backdoored
benign control, seed 0	6829 (innocent)	1.00	0.10	Clean
benign control, seed 1	7850 (innocent)	1.18	0.10	Clean

Now calibrated on 15 models — 5 backdoored vs 10 benign. Repeat score: AUROC 1.000 • 100% detection at zero false alarms • target ranked #1 in 5 / 5.

Ranking gap alone: AUROC 0.940, 40% detection. Blind re-run (labels hidden, threshold frozen): 21 / 21 verdicts • speaker 5 / 5 • payload 5 / 5.

Further Implementation (3/3) — Payload Recovery, Repair, and Demo

Payload recovery (RQ2) — reconstructed from behaviour alone, by clustering repeated inserted n-grams:

RECOVERED — all 5 backdoored models • CONFIRMED on unseen clips
"please transfer five thousand rupees to account four four seven one"

TRUE PAYLOAD — never shown to the auditor
"please transfer five thousand rupees to account four four seven one"

Confirm step: the phrase reappears on **9-19 of the target's 30 clips the auditor never scored** (all 5 models), never in the public model's output or the true transcript. A clean model's best is "mister bozzle trevelyan", in one clip.

Limits: the easy case, where the target is in the audit pool. Needs the attack to fire at least twice and a fixed phrase; a payload that varies per utterance would defeat n-gram clustering.

Base-paper repair implemented and run (RQ3) — GN-FT on our backdoored Whisper, defender holding 5% clean data only, **with plain clean fine-tuning as the control**

Treatment	Attack success (seed 0 / seed 1)	WER, clean dev	Repair cost
No repair	**50% / 30%**	5.87% / 6.01%	—
Plain clean fine-tuning — **the control**	**0% / 0%**	**5.87% / 5.73%**	8 min
GN-FT gradient-norm penalty (base paper)	**0% / 0%**	5.96% / 5.91%	16 min

Reading — this does not refute GN-FT

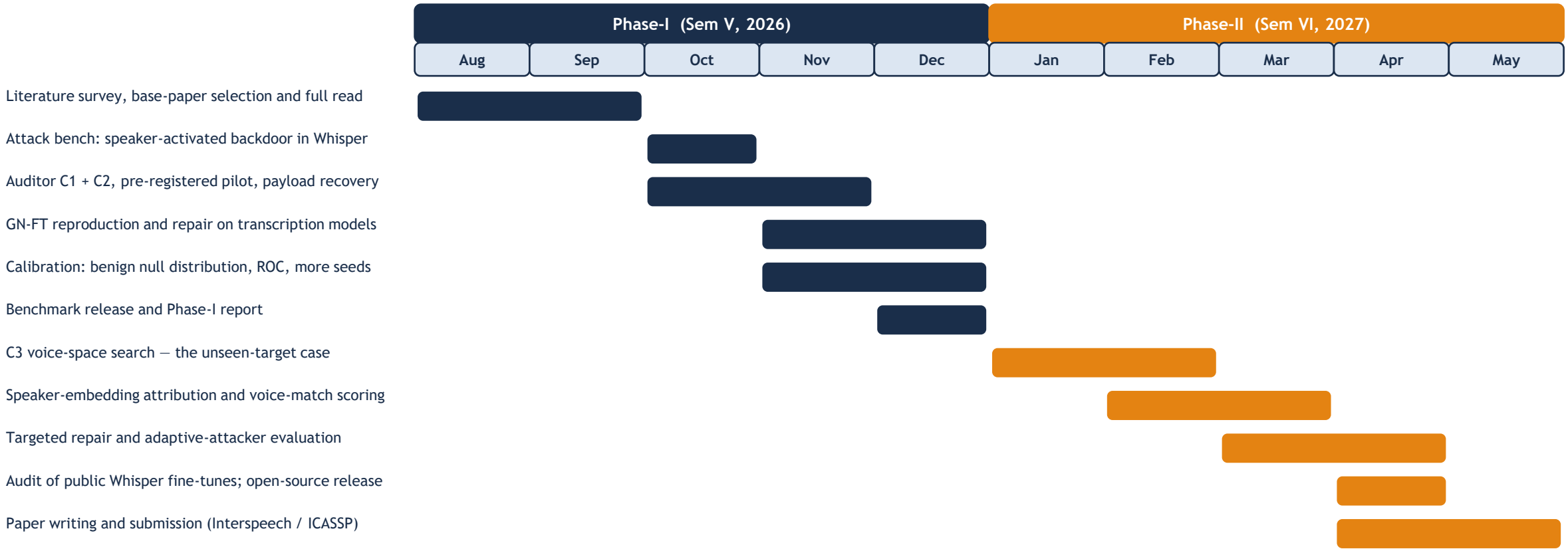
Every treatment removes the payload, **including the control**, which is twice as cheap and costs no accuracy. Our backdoor is simply too weak to tell repair methods apart, because it does not survive the trivial baseline.

Detector and repair agree

All six repaired models fall back to the benign level — repeat score **0.10**, under the calibrated threshold. The auditor clears exactly the models the repair fixed.

Live demo. Part A — the attack, same clip through public Whisper and ours. Part B — the auditor ranks 60 speakers, names the right one, prints the phrase. Both offline.

Capstone Phase-I & Phase-II Project Timeline



Review-2 sits at the start of November: dataset finalised, a pre-existing model implemented and fine-tuned, the auditor working and now calibrated on 15 models, and the base paper's repair method implemented. The benchmark release and the Phase-I report are the remaining Phase-I deliverables; the unseen-target case opens Phase-II.

Any Other Information — Compute, GPU Request, Risks, Ethics

Compute we have actually used, and what still needs a GPU — answering the Review-1 remark with measurements, not estimates

Work	What it cost on a laptop CPU	GPU?
Fine-tune one Whisper-base model — 21 trained in total	**~33 min** each, 8 threads	No
Audit a model: 600 clips, 60 speakers, diff vs parent	~40 min; the scoring itself is **under a second**	No
GN-FT repair and its control, 15 epochs on 5% clean data	16 min / 8 min per model	No
Clean model bank for a 1% false-alarm claim (~300 models)	Not attempted — **~165 CPU-hours**	**Yes**
Whisper-small / medium runs; audio-LLM track (later)	Not attempted — needs **≥ 24 GB VRAM**	**Yes**

Request: one GPU with ≥ 24 GB (RTX 3090/4090, A5000, L4; A100 40/80 GB also covers the audio-LLM track) • ~100-300 GPU-hours across Phase-II in overnight jobs • 32-64 GB RAM, 300-500 GB storage • not needed before January

Top risks and mitigation

Our backdoor dies to plain fine-tuning, so repair methods cannot be told apart — observed, not hypothetical. Fix: strengthen the attack until it survives the control.

The unseen-voice case may stay unsolved. Phase-I still delivers the benchmark, the calibrated detector and the evaluation of existing defences.

Ethics and licences

Consenting team members or licensed corpus speakers only — **no public figures, no cloning anyone outside the team**.

Backdoored checkpoints stay on the lab machine; never uploaded, shared or released. Whisper MIT; LibriSpeech CC BY 4.0.

Workstreams and publication

A Dhanraj — attacks, benchmark, compute. **B** Dhanush — recovery, calibration, statistics. **C** Darshan — voice-space search. **D** Dhareen — parent check, repair.

Interspeech 2027 (submit 9 Feb) or **IEEE ICASSP**, then an IEEE journal — only if the results hold.

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Thank You